

KTM_Camtrain_MBS

Model Package Guide

Architecture, features, modeling principles, validation, loads, excitation and reporting

Scope A consolidated engineering guide based on the developed front-bank, rear-bank, shared-crankshaft, measured cam-load and post-processing workflow.

Prepared for model users and developers

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Contents

1. Executive summary
 2. Package purpose and scope
 3. System architecture
 4. Component responsibilities and interfaces
 5. Kinematic and geometric foundations
 6. Compliant gear meshes and supported shafts
 7. External World, crankshaft and controller pattern
 8. Two-bank integration
 9. Measured camshaft loads
 10. Crankshaft speed irregularity and torsional excitation
 11. Initialization, numerical robustness and sign conventions
 12. Validation and commissioning workflow
 13. Recommended plots and acceptance checks
 14. Dymola and Python reporting workflow
 15. Package organization and naming
 16. Extension roadmap and design rules
- Appendix A. Key equations
- Appendix B. Quick-start checklist

Reading path New users should read Sections 1-4, then follow Section 12. Model developers should also read Sections 5-11 and 15-16.

1. Executive summary

KTM_Camtrain_MBS is a modular Modelica MultiBody package for building and validating compliant motorcycle camtrain models. The current baseline represents two camtrain banks driven by one externally supported crankshaft. Each bank contains a serial geartrain, supported shafts, compliant gear meshes, speed sensing and diagnostics. Optional measured, angle-dependent camshaft torques can be applied to all four camshafts.

- The physical environment is externalized through one inner MultiBody World.
- The crankshaft is a shared external component rather than duplicated inside either bank.
- The banks are passive mechanical branches connected to the same rotating crankshaft frame.
- One speed sensor and one actuator regulate the shared crankshaft.
- Measured cam loads are encapsulated in reusable components.
- Diagnostics cover speeds, ratios, residuals, mesh forces, powers and bank-to-bank differences.
- Dymola plot scripts support fast engineering review; Python can generate repeatable HTML reports.

Central principle Model boundaries should follow physical ownership. Shared equipment belongs at the wrapper level; bank-specific shafts and meshes remain inside the bank subsystem.

2. Package purpose and scope

The package supports staged development from unloaded kinematic validation to loaded, two-bank dynamic studies. It is aimed at understanding gear-train compatibility, compliant mesh response, camshaft load transmission, controller behavior and excitation sensitivity.

In scope	Typical use
Unloaded kinematics	Validate ratios, rotation signs and stable initialization.
Compliant dynamics	Assess mesh force, mesh deflection, damping and bearing motion.
Measured cam loads	Apply periodic torque tables using actual camshaft angle.
Two-bank interaction	Combine both bank reactions on one crankshaft.
Speed irregularity	Excite the camtrain with controlled order content.
Reporting	Create repeatable plots, metrics and HTML/PDF summaries.

The current model is a validation-oriented engineering model. It is not yet a complete combustion engine or full flexible cranktrain. Its modular interfaces are deliberately designed to allow those additions later.

3. System architecture

3.1 Top-level physical topology

World / common housing

- +-- External crankshaft bearings
- +-- Shared servo reaction
- +-- Front-bank housing
- +-- Rear-bank housing
- +-- Four cam-load reaction frames

External crankshaft rotating frame

- +-- FrontBank.driveIn
- +-- RearBank.driveIn
- +-- ServoTorque.frame_b
- +-- Crankshaft speed sensor

3.2 Signal topology

External crankshaft speed sensor

- +-- frontBank.crankshaftSpeedInput
- +-- rearBank.crankshaftSpeedInput
- +-- reporting diagnostics

Active PI controller

- > torqueCommand
- > shared servoTorque.torque[1]

3.3 Current ownership

Owner	Contains	Does not contain
Top-level wrapper	World, crankshaft, speed sensor, actuator, both banks, load components	Internal bank shafts and meshes
Bank subsystem	Four downstream shafts, four meshes, sensors, diagnostics	World, physical crankshaft, physical crank actuator
MeasuredCamLoad	Angle integration, lookup, activation, torque element, power	Bank geometry or controller
Rear-bank specialization	Rear geometry and controller default	Duplicated implementation

4. Component responsibilities and interfaces

Interface	Type	Meaning
frame_housing	MultiBody Frame_a	Common stationary engine-housing reference.
driveIn	MultiBody Frame_a	Rotating crankshaft input to the first gear mesh.
frame_intakeCamshaft	MultiBody Frame_b	External attachment point for intake-cam load.
frame_exhaustCamshaft	MultiBody Frame_b	External attachment point for exhaust-cam load.
crankshaftSpeedInput	RealInput [rad/s]	Measured speed of the shared external crankshaft.
torqueCommand	RealOutput [N.m]	Torque request generated by the enabled controller.
camshaftSpeed	RealInput [rad/s]	Actual speed passed into a measured cam-load component.
activation	RealInput	Smooth load activation factor.

Mechanical and signal interfaces Frame connectors transmit position, orientation, force and torque. RealInput and RealOutput connectors transmit numerical signals only. Keep these roles separate.

4.1 Connection-set principle

Modelica connections are transitive. All stationary frames connected to the housing form one connection set; all components connected to the external crankshaft frame form another. The stationary and rotating sets must never be accidentally joined.

5. Kinematic and geometric foundations

5.1 Shaft-center geometry

Each bank uses a shaft-position matrix. Row 1 is the crankshaft location even though the crankshaft is external. Rows 2-5 locate the step shaft, idler-cam shaft, intake camshaft and exhaust camshaft. Mesh center vectors are derived by subtraction, ensuring one source of geometric truth.

```
rCrankToStep    = shaftPos[2,:] - shaftPos[1,:]
rStepToIdlerCam = shaftPos[3,:] - shaftPos[2,:]
rIdlerCamToIntake = shaftPos[4,:] - shaftPos[3,:]
rIntakeToExhaust = shaftPos[5,:] - shaftPos[4,:]
```

5.2 Gear ratios and expected speeds

Stage	Ratio magnitude	Expected sign
Crank to step	24/40 = 0.6	Reversal
Step to idler-cam	25/51 = 0.490196	Reversal
Idler-cam to intake	51/30 = 1.7	Reversal
Intake to exhaust	30/30 = 1.0	Reversal
Crank to intake	0.5	Three reversals
Crank to exhaust	0.5	Four reversals

At 6000 rpm Expected signed speeds are +6000, -3600, +1764.71, -3000 and +3000 rpm from crankshaft through exhaust camshaft.

5.3 Rear-bank geometry

The current rear-bank example mirrors the front-bank y coordinates. This is a configuration assumption, not a universal rule. Replace the coordinates with measured or CAD-derived rear-bank geometry when available. Both banks must reference the same physical crankshaft location.

6. Compliant gear meshes and supported shafts

6.1 Supported shaft model

Each downstream shaft uses ShaftRadialBush_Base with two radial support elements and a shaft body. Housing position locates the support system, while shaftLocation and bearingDistance define the internal axial arrangement. Bearing stiffness and damping are parameterized at bank level.

- Connect both bearing-support frames to frame_housing.
- Connect the rotating shaft frame to meshes, sensors and external cam-load interfaces.
- Do not duplicate a physical shaft when exposing it at a subsystem boundary.
- Preserve validated shaft parameterization during refactoring.

6.2 Compliant gear mesh

CompliantSpurGearMesh represents mesh stiffness, damping, ratio and alignment. It exposes mesh force, deflection and damping power for validation. Moderate stiffness should be used first; increase stiffness only after kinematics and initialization are stable.

Parameter	Baseline	Role
meshStiffness	1e6 N/m	Linearized mesh stiffness
meshDamping	20 N.s/m	Linearized mesh damping
initialDeflection	0	Initial mesh compression
transmissionErrorAmplitude	0	Disabled during baseline validation
useUnilateralContact	false	Bilateral baseline contact

7. External World, crankshaft and controller pattern

7.1 Why externalize?

- One World serves the complete hierarchy.
- One crankshaft can drive multiple bank subsystems.
- One actuator sees the combined mechanical reaction.
- The bank becomes reusable with other cranktrain models.
- The model boundary becomes compatible with future torsional crankshaft models.

7.2 External crankshaft migration pattern

Inside bank:
remove internal crankshaft
connect driveIn -> meshCrankToStep.frame_a
replace crank sensor by crankshaftSpeedInput

Inside wrapper:
instantiate crankshaft
connect crankshaft bearings -> common housing
connect crankshaft.frame_b -> bank.driveIn
connect speed sensor -> crankshaft.frame_b
feed measured speed -> bank crankshaftSpeedInput

7.3 Transitional controller arrangement

The current two-bank model enables the front-bank PI controller and disables the rear-bank controller. This avoids two integrators controlling one actuator. Both banks still receive the same crankshaft speed for ratio and residual diagnostics.

Long-term recommendation Move the single mean-speed controller to the top-level wrapper. The bank subsystems should ultimately be passive mechanical plants.

8. Two-bank integration

Both bank drive inputs connect to the same external crankshaft frame. Modelica automatically sums their cut forces and torques in the shared connection set. No manual summation of bank reaction torques is required.

```
connect(ext_crankshaft.frame_b, frontBank.driveIn);
connect(ext_crankshaft.frame_b, rearBank.driveIn);
connect(externalCrankshaftSpeedSensor.w[1], frontBank.crankshaftSpeedInput);
connect(externalCrankshaftSpeedSensor.w[1], rearBank.crankshaftSpeedInput);
```

8.1 Rules for adding banks

- Use one shared crankshaft position and orientation reference.
- Give each bank independent shaft geometry and load phase parameters.
- Use only one active controller for the shared actuator.
- Keep bank comparison diagnostics for symmetric unloaded tests.
- Introduce axial offsets or torsional crankshaft segments if the physical bank drives are not co-located.

9. Measured camshaft loads

9.1 Reusable MeasuredCamLoad

Each load component integrates actual camshaft speed to obtain a continuous angle, evaluates a measured torque table, scales and activates the result, then applies the torque between housing and camshaft. Encapsulation prevents four duplicated processing chains in the wrapper.

```
scaledTorque = torqueScale * rawTorque
appliedTorque = loadApplicationSign * activation * scaledTorque
loadPower = appliedTorque * camshaftSpeed
```

9.2 Independent sign conventions

Parameter	Purpose	Do not confuse with
angleSign	Maps physical rotation to table-angle direction	Mechanical torque direction
tableTorqueSign	Corrects sign stored in data file	Camshaft rotation direction
loadApplicationSign	Defines physical applied-torque direction	Torque-file convention
phaseOffset	Aligns table waveform with cam angle	Mechanical gear position

9.3 Four-load structure

- frontIntakeLoad
- frontExhaustLoad
- rearIntakeLoad
- rearExhaustLoad

A common activation ramp is useful during commissioning. Separate ramps can be introduced later. Start every load scale at zero, then activate one path at a time.

9.4 Torque reflection

$$T_{crank} = T_{cam} * \omega_{cam} / \omega_{crank}$$

This is a power-equivalent diagnostic. Do not feed it automatically into the actuator unless intentionally implementing feedforward control.

10. Crankshaft speed irregularity and torsional excitation

10.1 Controlled speed-reference excitation

$$\text{omega_target}(t) = \text{omega_mean} + \text{sum}(\text{DeltaOmega_i} * \sin(\text{order_i} * \text{theta_ref} + \text{phase_i}))$$

This mode prescribes the desired speed waveform through the controller. It is suitable for controlled transfer studies, order sweeps and repeatable comparison of camtrain response.

10.2 Physical torque excitation

$$T_crank(t) = T_PI(t) + \text{sum}(\text{DeltaT_i} * \sin(\text{order_i} * \text{theta_crank} + \text{phase_i}))$$

This is the more physical representation of gas-pressure and firing-order excitation. A slow controller maintains mean speed, while a separate torque source excites the crankshaft. The resulting speed irregularity emerges from inertia, stiffness, damping and load interaction.

10.3 Engine-order interpretation

Mean speed	Crank frequency	Order 0.5	Order 1	Order 2
6000 rpm	100 Hz	50 Hz	100 Hz	200 Hz

Controller bandwidth A high-bandwidth speed controller can suppress the torsional irregularity being studied. Use a slow mean-speed controller for physical torque excitation.

11. Initialization, numerical robustness and sign conventions

11.1 Consistent initial speeds

Downstream shaft start speeds are derived from the external crankshaft reference and gear ratios. The wrapper must pass the same reference speed into both banks. This avoids mesh shock caused by inconsistent initial motion.

11.2 Saturation and anti-windup

```
actuatorTorque = limit(actuatorTorqueUnlimited, +/- actuatorTorqueLimit)
der(speedErrorIntegral) = 0 when disabled or saturated
```

The disabled rear controller remains structurally complete: its unlimited and limited torque are zero and its integral state is frozen.

11.3 Structural singularity checklist

- Every RealInput must be connected or assigned exactly once.
- Every Torque.torque component must be supplied exactly once.
- Do not define one diagnostic variable with both nominal-ratio and actual-speed equations.
- Reconnect every retained MultiBody sensor to its shaft frame.
- Remove self-connections and obsolete inherited interfaces.
- Use component-qualified parameter names only when intentionally referencing another instance.

11.4 Numerical staging

- Start unloaded and with moderate mesh stiffness.
- Use bilateral contact initially.
- Ramp loads smoothly.
- Increase torque scale, stiffness and model detail one change at a time.
- Preserve a known-good baseline result for comparisons.

12. Validation and commissioning workflow

Stage	Configuration	Primary acceptance question
1	Single shaft	Validate supported-shaft acceleration under known torque.
2	Single bank, internal crankshaft	Validate signs, ratios and compliant meshes.
3	External World and actuator	Verify unchanged flattened topology.
4	External crankshaft	Verify driveIn, sensor feedback and controller closure.
5	Two banks, unloaded	Confirm common crankshaft and bounded first-stage responses.
6	One cam load	Validate angle, torque and power conventions.
7	Four cam loads	Activate individually, then together.
8	Speed irregularity	Use one small order first; expand harmonic set.
9	Physical torque excitation	Introduce separate gas/firing torque source.
10	Sensitivity and reporting	Automate parameter sweeps and acceptance metrics.

12.1 Loaded simulation duration

With a load ramp starting at 0.05 s and lasting 0.10 s, full activation occurs at 0.15 s. A stop time of 0.3 s provides approximately 15 cam revolutions at 3000 rpm and supports cycle-based assessment.

13. Recommended plots and acceptance checks

Plot group	Variables / intent
Crankshaft control	reference and measured rpm; mean and instantaneous error; actuator torque; saturation
Signed speeds	crankshaft plus all four shafts in both banks
Ratios	all stage and total actual ratios for each bank
Compatibility	six signed residuals per bank
Mesh response	force and deflection for all eight meshes
Cam loads	angle, raw torque, scaled torque, applied torque and power for four loads
Power	actuator, bank load powers, mesh damping, reflected torque
Bank comparison	step, idler, intake and exhaust speed differences
Bearing motion	radial displacement and velocity at both supports of each shaft

13.1 Core acceptance criteria

- Mean crankshaft speed remains close to target.
- The actuator does not remain continuously saturated.
- Signed speeds and ratios agree with physical topology.
- Residuals, mesh forces, mesh deflections and bearing motions remain bounded.
- Load activation is smooth and produces no nonphysical discontinuity.
- Average load power has a physically plausible sign.
- No unexplained drift appears in angle-independent diagnostics.
- Front/rear differences are small for symmetric unloaded configurations.

14. Dymola and Python reporting workflow

14.1 Dymola plot scripts

Create focused plot windows interactively, generate a reusable Dymola plot script and store it under Resources/Scripts/Dymola. Use a compact overview script automatically after simulation, and keep detailed scripts manually callable.

14.2 Python HTML reports

A Python report generator can read Dymola MAT files, compute metrics over a defined evaluation window, render plots and assemble a self-contained HTML report. Isolate MAT-reader calls behind a small adapter so the reader library can be replaced later.

Report level	Purpose	Typical contents
Quick check	Immediate engineering sanity	Control, first meshes, applied loads, power
Validation report	Formal assessment of one run	Metrics, limits, all plots, metadata, conclusions
Comparison report	Compare configurations	Overlaid runs, deltas, trend tables, pass/fail

14.3 Reproducibility metadata

- Model and result-file names
- Dymola and Modelica versions
- library or Git revision
- solver, tolerance and simulation time
- mean speed and excitation orders
- controller limits and gains
- bearing and mesh parameters
- load files, phase offsets, signs and scales
- evaluation-window definition

15. Package organization and naming

```
KTM_Camtrain_MBS/  
  Parts/  
    ShaftRadialBush_Base  
    CompliantSpurGearMesh  
  Components/  
    MeasuredCamLoad  
  Sources/  
    MeasuredCamTorque1D  
    CrankshaftSpeedIrregularity  
  Examples/Banks/  
    FrontBankKinematics_NoWorld_noCS  
    RearBankKinematics_NoWorld_noCS  
    FrontBankKinematics_Test_TwoBanks  
  Resources/  
    Data/camTorque/  
    Scripts/Dymola/  
    Reports/  
  python/  
    build_dymola_report.py
```

15.1 Naming guidance

- Use physical names for frame connectors, such as driveIn and frame_intakeCamshaft.
- Use NoWorld only for subsystems intentionally requiring an enclosing inner World.
- Avoid names like World for test wrappers because they can be confused with MultiBody.World.
- Distinguish test, validation and reusable plant models.
- Prefer lower camel case for component instances and upper camel case for classes.

16. Extension roadmap and design rules

16.1 Recommended next architectural step

Move the single PI controller from the front-bank subsystem to the top-level wrapper. Both banks then become passive mechanical plants, simplifying reuse with alternative crankshaft excitation and cranktrain models.

16.2 Potential extensions

- Flexible or segmented crankshaft with torsional compliance between bank drives.
- Measured combustion or firing-order torque maps.
- Gear transmission error and unilateral contact.
- Physical intake and exhaust cam torque files for both banks.
- CAD-derived rear-bank geometry and phase alignment.
- Bearing nonlinearities and clearances.
- Parameter records for geometry, bearings, meshes and controls.
- Automated frequency/order sweeps and sensitivity analysis.
- Cycle-synchronous statistics and order tracking in Python.
- Regression tests with numerical acceptance thresholds.

16.3 Durable model-building rules

Rule 1 One physical component should exist once. Expose it through connectors instead of duplicating it.

Rule 2 Keep mechanical excitation, control logic and diagnostics conceptually separate.

Rule 3 Parameterize geometry once and derive vectors and ratios from it.

Rule 4 Make sign conventions explicit and independently testable.

Rule 5 Refactor in validated increments, preserving a baseline after each successful step.

Appendix A. Key equations

Quantity	Equation / definition
Angular speed from rpm	$\omega = \text{rpm} * 2\pi / 60$
Stage ratio	downstream speed magnitude / upstream speed magnitude
Compatibility residual	$\omega_{\text{downstream}} + \text{ratio} * \omega_{\text{upstream}}$ for an external gear pair
PI command	$T_{\text{unlimited}} = K_p * \text{speedError} + K_i * \text{integralError}$
Cam load	$T_{\text{applied}} = \text{sign} * \text{activation} * \text{scale} * T_{\text{table}}$
Load power	$P_{\text{load}} = T_{\text{applied}} * \omega_{\text{cam}}$
Reflected crank torque	$T_{\text{crank}} = T_{\text{cam}} * \omega_{\text{cam}} / \omega_{\text{crank}}$
Order frequency	$f_{\text{order}} = \text{order} * \text{rpm}_{\text{mean}} / 60$
Speed irregularity	$\omega_{\text{target}} = \omega_{\text{mean}} + \text{sum of order harmonics}$

Appendix B. Quick-start checklist

- ☐ Confirm one inner World at the top level.
- ☐ Confirm one external crankshaft and one actuator.
- ☐ Confirm both banks connect to the same crankshaft frame.
- ☐ Confirm both banks receive the same crankshaft speed signal.
- ☐ Confirm only one controller is enabled.
- ☐ Set all four cam-load scales to zero for the first run.
- ☐ Check signed shaft speeds and nominal ratios.
- ☐ Check first-stage residuals, forces and deflections.
- ☐ Activate one cam load at a time with a small scale.
- ☐ Run long enough to reach full activation.
- ☐ Check actuator saturation and power signs.
- ☐ Save a result MAT file and generate a repeatable report.

Document status This guide summarizes the working development baseline discussed through September 2026. Update geometry, torque files, phase conventions and acceptance thresholds as validated physical data become available.